

Payments and credits

How money works in FunDive. Rules the code depends on; the reasoning behind each one lives in comments at the site that implements it.

FunDive moves no money. There is no payment processor. Bank transfers, cash and card happen off-app and are recorded by hand. Account credit is the one exception — it is internal, and the only "payment" the app creates itself.

Two ledgers

Both append-only. Nothing is ever edited in place: a refund is a new `payments` row, a discount a new `booking_amendments` row.

Table	Holds
<code>payments</code>	Money received. <code>pending</code> / <code>paid</code> / <code>refunded</code> / <code>voided</code> .
<code>credits</code>	Money owed back to a diver. <code>open</code> or <code>settled</code> .

Alongside them, `bookings.details` carries the `total` and deposit **frozen at booking time**, so later price changes never alter what someone was quoted.

What a booking owes

```
owed      = details.total +  $\Sigma$  booking_amendments.amount
paid      =  $\Sigma$  payments: 'paid' adds, 'refunded' subtracts, rest ignored
credit    =  $\Sigma$  open credits tied to THIS booking
balance   = owed - paid - credit
depositDue = max(0, min(details.deposit, owed) - paid)
```

`netPaid()` and `booking_net_paid()` are the only implementations of that signing rule — every screen and every RPC must use one, or they disagree about a partly refunded booking.

The deposit is **clamped to owed**, because a discount can push the frozen deposit above the remaining balance. Covering it promotes a `pending` booking to `confirmed`.

A negative balance is **credit to the diver**, whether it came from an awarded credit or from simply overpaying. There is no separate "overpaid" concept.

Credits

Every credit records a `source`: `manual`, `event_cancellation`, `booking_cancellation_return`, or `carry_forward`. Only the middle two mean *this booking's money has been given back*, and only they block a further automatic refund — a goodwill award tied to the same booking must not.

Account credit — the balance a diver sees — is *open credits not tied to an active booking, plus every overpayment*. So a credit tied to a cancelled booking is spendable again, and bookings a lead booker pays for count toward the lead, not the diver.

Spending credit

One RPC, `apply_credit_to_booking`, because divers cannot write `payments` or `credits` directly. A diver may spend against their own booking, a parent against their child's, an admin against anyone's — and the credit spent is always the booking owner's.

It clamps to what is owed and what is available, consumes credit oldest-first, carries any remainder forward as a new `carry_forward` row, records an offsetting `account_credit` payment, and confirms the booking if that covers the deposit. It refuses a **cancelled** booking outright: that trip will never happen, so credit spent there would be destroyed.

Cancellations

Four of them, and only one returns money by itself.

What happened	Money returned automatically?
Shop cancels the event	Yes — every registrant gets credit worth their full net paid
Refund request approved	No
Admin cancels one booking	No
Diver cancels their own booking	No

In the bottom three a trigger does return any **account credit** spent on the booking — that money never left the app. Cash and transfers did, so only a person can move them back.

Cancelling an event does **not** cancel its bookings. The credit is what settles things with the diver. Issuing runs after the cancel commits, so a failure cannot un-cancel the event.

A diver may self-cancel only a `pending` booking with nothing paid and no refund request outstanding (`canSelfCancel`). This is a product rule, not a security boundary — the database lets a diver cancel any of their own bookings — so it must be applied at **every** diver-facing cancel control.

Money left on a cancelled booking

A cancelled booking's balance short-circuits to **settled**. That means *nothing further is owed for this trip*, not *the money has been dealt with*. Its frozen total is not a debt, and netting it would double-count any cancellation credit.

So when a diver who paid cash cancels, the shop still holds their money and no other screen shows it. Exactly two endings are correct — record a `refunded` payment, or issue a credit — and doing both pays them twice.

`/admin/refunds` lists these under **Cancelled bookings still holding money**, with a button for each ending. It is the only surface that can see them: balances read settled, account credit skips cancelled bookings, and the refund queue lists only non-cancelled ones.

Reporting

Applied account credit is **not revenue**. The cash arrived earlier, on the booking that generated the credit; counting it again books the same money twice. `isExternalPayment()` guards every cash-revenue sum, and credit spent is reported beside cash, never inside it.

Balance math is the opposite and counts it in full — it really did settle that booking. Staff revenue is different again: $\text{base price} \times \text{confirmed heads}$, deliberately not what has been banked.

Rules that must not break

1. Never edit money rows in place. New row, always.
2. Sum paid through `netPaid / booking_net_paid`. Nowhere else.
3. `owed` is `details.total` **plus amendments**, in TypeScript and SQL.
4. Clamp the deposit to `owed` wherever you compare against it.
5. A booking's own credit is never spendable against that booking.
6. A cancelled booking has no live balance and refuses credit.
7. `diverCreditBalance` is the only definition of account credit.
8. Every automatic credit records its `source`.
9. Cash-revenue sums apply `isExternalPayment`. Balance math does not.
10. Constraints, triggers and RLS policies get integration tests.

Known gaps

- A confirmation PDF's credit line is a prediction: credit is applied after the booking lands, but the PDF renders before that.
- `credits.currency / payments.currency` default to `TWD`. Every writer names a currency explicitly except a group payment recording the first money on a booking. Forks on another currency should change the defaults in their own baseline.
- Restoring a cancelled event leaves its credits issued.
- Keeping a cancellation fee has no dedicated action, so the booking stays on the holding list.